Tongyu Nie

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Research Interests

My research interests include Virtual and Augmented Reality, Human-Computer Interaction, and Visual Perception. My dissertation focuses on the design and evaluation of cybersickness mitigation techniques.

Education

2020 – 2026 (Expected) **Ph.D. in Computer Science**, *University of Minnesota*, Minneapolis, MN, USA.
Advisor: Evan Suma Rosenberg. Date of candidacy: Jan 3, 2023.

2019 – 2025 **M.Sc. in Computer Science**, *University of Minnesota*, Minneapolis, MN, USA.

2014 – 2018 **B.Eng. in Information Security**, *Harbin Institute of Technology*, Weihai, Shandong, China.

2017 **Visiting International Student**, *University of Wisconsin-Madison*, Madison, WI, USA.

Awards and Honors

2026 Camp Continuum for Early Career XR Scholars, Camper (Sponsored by IEEE VGTC)

2025 Doctoral Dissertation Fellowship, University of Minnesota, $25,000

2025 Best Paper Award (top 1% of total submissions), IEEE Conference on Virtual Reality and 3D User Interfaces (VR)

Publications

∗ Asterisk denotes equal contribution.

In Computer Science, conference papers undergo full peer review and are generally considered equivalent in prestige to journal articles in other fields. The top conferences in virtual reality (IEEE VR and IEEE ISMAR) implement a unified submission and review process of journal and conference papers, with higher-scoring submissions being recommended for the IEEE Transactions on Visualization and Computer Graphics (TVCG) journal.

Journal Papers

J3 **Tongyu Nie**, Yuxuan Huang, Ville Cantory, Danhua Zhang, and Evan Suma Rosenberg. Effects of field-of-view restriction and peripheral teleportation on path integration during virtual locomotion. *IEEE Transactions on Visualization and Computer Graphics*, 2026. (Proc. IEEE ISMAR 2026, Acceptance Rate: 15.9%).

J2 **Tongyu Nie***, Ville Cantory*, Courtney Hutton Pospick*, Danhua Zhang, Haoyu Tan, Sam Adeniyi, Fei Wu, Daniel Zielasko, Isayas Berhe Adhanom, and Evan Suma Rosenberg. Revenge of the Sick: A Meta-Analysis of Washout Periods in Cybersickness Research. *IEEE Transactions on Visualization and Computer Graphics*, 2026. **[Corresponding Author]**.

J1 **Tongyu Nie**, Courtney Hutton Pospick, Ville Cantory, Danhua Zhang, Jasmine Joyce DeGuzman, Victoria Interrante, Isayas Berhe Adhanom, and Evan Suma Rosenberg. Peripheral Teleportation: A Rest Frame Design to Mitigate Cybersickness During Virtual Locomotion. *IEEE Transactions on Visualization and Computer Graphics*, 2025. (Proc. IEEE VR 2025, Acceptance Rate: 17.3%.) **[Best Paper Award, Top 1%]**.

Conference Papers

C6 Xianhao Carton Liu, Difan Jia, **Tongyu Nie**, Evan Suma Rosenberg, Victoria Interrante, and Chen Zhu-Tian. Can AR Embedded Visualizations Foster Appropriate Reliance on AI in Spatial Decision-Making? A Comparative Study of AR X-Ray vs. 2D Minimap. In *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems*, 2026. Acceptance Rate: 25.3%.

C5 Yuxuan Huang, Qiao Jin, **Tongyu Nie**, Victoria Interrante, and Evan Suma Rosenberg. Context when needed, digits when useful: Balancing memorability and efficiency in virtual reality authentication. In *Proceedings of the 2026 32nd ACM Symposium on Virtual Reality Software and Technology (Accepted)*, 2026. Acceptance Rate: 27.0%.

C4 Jasmine Joyce DeGuzman, Kaori Hirano, Tabitha Peck, Alice Guth, Evan Suma Rosenberg, and **Tongyu Nie**. Reduction of Motion Complexity as an Objective Indicator of Cybersickness in Virtual Reality. In *2025 IEEE Conference Virtual Reality and 3D User Interfaces (VR)*, pages 156–164, 2025. Acceptance Rate: 26.9%.

C3 Ville Cantory, Darya Biparva, Haoyu Tan, **Tongyu Nie**, John Schroeder, Ruofei Du, Victoria Interrante, and Piotr Didyk. Enhancing Foveated Rendering with Weighted Reservoir Sampling. In *The 18th ACM SIGGRAPH Conference on Motion, Interaction, and Games (MIG '25)*, pages 2–12, 2025. Acceptance Rate: 41.3%.

C2 Haoyu Tan, **Tongyu Nie**, and Evan Suma Rosenberg. Invisible Mesh: Effects of X-Ray Vision Metaphors on Depth Perception in Optical-See-Through Augmented Reality. In *2024 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*, pages 376–386, 2024. Acceptance Rate: 28.7%.

C1 **Tongyu Nie**, Isayas Berhe Adhanom, and Evan Suma Rosenberg. Like a Rolling Stone: Effects of Space Deformation During Linear Acceleration on Slope Perception and Cybersickness. In *2023 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*, pages 658–668, 2023. Acceptance Rate: 21.2%.

Extended Abstracts

EA3 Jasmine Joyce DeGuzman, Erik DeVries Smith, Samyok Nepal, Kalinda Miller, Courtney Hutton Pospick, **Tongyu Nie**, and Evan Suma Rosenberg. Walk Me Through It: Using Impossible Spaces to Embody Graph Traversal Algorithms. In *2024 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*, pages 1092–1093, 2024.

EA2 Catherine Borisova, Ruth Mesfin, **Tongyu Nie**, and Evan Suma Rosenberg. Build Your Own Trigonometric Adventure in Virtual Reality. In *2024 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*, pages 1094–1095, 2024.

EA1 **Tongyu Nie** and Evan Suma Rosenberg. Redirected Tilting: Eliciting Postural Changes with a Rotational Self-Motion Illusion. In *2021 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*, pages 178–182, 2021.

Research Experience

08/2020 – Present **Graduate Research Assistant**, *Illusioneering Lab*, University of Minnesota.

- Developed two novel cybersickness mitigation techniques using shaders in HLSL in collaboration with multi-disciplinary researchers in Kinesiology.
- Explored the use of geometry deformation on cybersickness and uneven terrain simulation.
- Designed a rest frame using content from the same virtual environment users were immersed in and achieved state-of-the-art results in cybersickness mitigation.
- Evaluated the impact of mitigation techniques on visual path integration.
- Conducted a meta-analysis to provide guidance for cybersickness experimental design.
- Mentored multiple undergraduate and graduate students, leading to publications in premier venues.

Summer 2024–2026 **NSF REU Mentor**, *REU Site: Human-Centered Computing for Social Good*, University of Minnesota.

- Supervised 10+ undergraduate students from underrepresented groups during their summer research experiences; served as a point of contact and intermediary between students, the department, and supervising professors.
- Helped students on individual projects in graphics, visualization, virtual reality, robotics, and human-computer interaction; wrote and taught 8 weekly modules covering the research process, data analysis, and scientific writing.
- Organized a graduate student panel on graduate school pathways.

05 – 08/2021 **Research Intern**, *Institute for Creative Technologies*, University of Southern California.

- Designed novel uncertainty visualization metaphors for augmented reality display.
- Conducted an online human-subject study to evaluate the effectiveness of visualization techniques.

Mentoring Experience

Master's Students

2022 – 2024 Haoyu Tan (Now: PhD at UMN)

Undergraduate Students

2026 Angela Enciso

2025 Meryem Koksal

2021 – 2025 Jasmine Joyce DeGuzman (Now: PhD at UCF)

2024 Kaori Hirano

2024 Shauntavia Hooper

2023 – 2024 Catherine Borisova

2023 – 2024 Ruth Mesfin

2022 Christian Halvorson

2022 Caleb Calomarde

Teaching Experience

Graduate Teaching Assistant, *University of Minnesota*, Duties included leading labs, holding office hours, grading, and developing course materials.

Fall, 2026 CSCI 3081W: Program Design and Development

Spring, 2025 CSCI 1933: Introduction to Algorithms and Data Structures

Fall, 2024 CSCI 1933: Introduction to Algorithms and Data Structures

Fall, 2023 CSCI 5619: Virtual Reality and 3D Interaction

Spring, 2022 CSCI 5552: Sensing and Estimation in Robotics

Fall, 2021 CSCI 1913: Introduction to Algorithms, Data Structures, and Program Development

Spring, 2021 CSCI 1913: Introduction to Algorithms, Data Structures, and Program Development

Fall, 2020 CSCI 5619: Virtual Reality and 3D Interaction

Academic Activities

Workshop Organization

2027 IEEE VR Workshop on Immersive Sickness Prevention (WISP) Lead organizer

2026 IEEE VR Workshop on Immersive Sickness Prevention (WISP) Co-organizer

Program Committee Service

2027 IEEE VR 2027 Conference Program Committee

2025 – 2026 EuroXR Scientific International Program Committee

2026 IEEE ISMAR 2026 Poster Program Committee

2026 IEEE VR 2026 Poster Program Committee

2025 IEEE VR 2025 Poster Program Committee

2021 – 2025 IEEE VR Workshop on Immersive Sickness Prevention (WISP) Program Committee

Journal & Conference Reviewing

2023 – 2027	IEEE Conference on Virtual Reality and 3D User Interfaces (VR)	*30 reviews*
2023 – 2026	IEEE International Symposium on Mixed and Augmented Reality (ISMAR)	*20 reviews*
2024 – 2025	IEEE Visualization Conference (VIS)	*3 reviews*
2025	IEEE Pacific Visualization Conference (PacificVis)	*2 reviews*
2024 – 2027	ACM CHI Conference on Human Factors in Computing Systems (CHI)	*11 reviews*
2026	ACM Symposium on User Interface Software and Technology (UIST)	*1 review*
2025	ACM Conference on Computer-Supported Cooperative Work (CSCW)	*1 review*
2023 – 2026	ACM Symposium on Virtual Reality Software and Technology (VRST)	*10 reviews*
2024 – 2025	ACM Symposium on Spatial User Interaction (SUI)	*5 reviews*
2023 – 2026	IEEE Transactions on Visualization and Computer Graphics (TVCG)	*5 reviews*

Special Recognitions.

CHI 2025 Special Recognitions for Outstanding Reviews

Academic Service

Student Volunteer.

IEEE VR 2020, IEEE VR 2024, IEEE ISMAR 2024

University/Department Service.

2024 Human Centered Computing Seminar: Student Coordinator

2022 Council of Graduate Students Grants Review Committee: Reviewer

2021 Empowering Women in Science – A Friend in STEM Mentoring Program: Mentor

Invited Presentation

Invited Talk.

- IEEE ISMAR Invited Journal Paper Talk. "Revenge of the Sick: A Meta-Analysis of Washout Periods in Cybersickness Research" IEEE ISMAR 2026 (Bari, Italy). Oct 8, 2026.
- Extended Horizon Lab Meeting Invited Talk. "Design and Evaluation of Cybersickness Mitigation Techniques in Virtual Reality" North Carolina State University (Raleigh, NC). Feb 17, 2026.
- Workshop on Immersive Sickness Prevention. "The Influence of Carryover Effects on Cybersickness Experiment Outcomes: A Meta-Analysis" IEEE VR 2025 (Saint-Malo, France). Mar 8, 2025.
- Workshop on Locomotion and Wayfinding in XR. "Peripheral Teleportation: A Rest Frame Design to Mitigate Cybersickness During Virtual Locomotion." IEEE VR 2025 (Saint-Malo, France). Mar 8, 2025.
- Virtual Experiences Research Group Lab Meeting Invited Talk. "Advancing Cybersickness Mitigation in Virtual Reality" University of Florida (Gainesville, FL). Feb 17, 2025.
- Workshop on Immersive Sickness Prevention. "Peripheral Teleportation: A Rest Frame Design to Mitigate Cybersickness." IEEE VR 2024 (Orlando, FL). Mar 16, 2024.

Guest Lecture.

- CSCI 5619: VR and 3D Interaction "Cybersickness in Virtual Reality." University of Minnesota. Nov 14, 2025.
- CSC 591 (008): XR and 3D Interaction "Cybersickness in Virtual Reality." North Carolina State University. Oct 30, 2025.
- CSCI 8001: Intro to Research in CS, I "Grad Student Panel." University of Minnesota. Sept 19, 2025.
- CSCI 1933: Intro to Algorithms & Data Structures "Exceptions and Iterators in Java." University of Minnesota. March 28, 2025
- CSCI 5619: VR and 3D Interaction "Motion Filtering." University of Minnesota. Oct 13, 2023.
- CSCI 1913: Intro to Algorithms & Program Development "Version Control System and Git." University of Minnesota. Apr 15, 2021.